[INSERT REPORT TITLE]

Preliminary Working Report — Internal Draft

K. Vishnu Kumar Adithya Kesan Jayakanth Adithya Anand

26 May 2026

[School Name] · Team ID: [INSERT TEAM ID]

NOT FOR SUBMISSION — distil into the 10-page template before 27 May 1500 SGT

Contents

1. Executive Summary

Three short paragraphs: (i) the problem; (ii) approach taken; (iii) headline findings. Same content as the 10-page version but written first here, so the distillation is easier.

Headline results.

- **T1.** ...
- **T2.** ...
- **T3.** ...
- **T4.** ...
- **T5.** ...
- **T6.** ...
- **T7.** ...
- **T8.** ...
- **Exploratory.** ...

2. Introduction

2.1 Background and context

Several paragraphs: the broader topic, recent developments, why the question matters.

2.2 Motivation

Why this challenge framing is interesting; the gap between the prescribed eight tasks and the underlying scientific / engineering question.

2.3 Scope of this report

What is and is not addressed. Mention items deliberately deferred to Future Work.

2.4 Structure of this report

Section-by-section map for readers; mirrors the table of contents.

3. Problem Statement

3.1 Verbatim challenge restatement

Copy the problem prompt; mark any ambiguities clarified in the Coursemology forum on 26 May 0930–1200.

3.2 Decomposition into the eight tasks

T1. *Task 1 — full statement.*

T2. *Task 2 — full statement.*

T3. *Task 3 — full statement.*

T4. *Task 4 — full statement.*

T5. *Task 5 — full statement.*

T6. *Task 6 — full statement.*

T7. *Task 7 — full statement.*

T8. *Task 8 — full statement.*

3.3 Success criteria

For each task, what counts as a complete answer (a number, a procedure, a recommendation, a proof, etc.). Useful for the team to align early.

4. Literature & Prior Art

4.1 Mathematical foundations

The mathematical objects this challenge sits on — graph theory, statistics, optimisation, dynamical systems, etc. — with the key references.

4.2 Comparable challenges and prior champion reports

What did the 2022 and 2024 champion reports do well? What patterns repeat across past SIMC problems? Cite `SIMC-Report-2022-Champion.pdf` and `SIMC2_0_Champion_Report_2024.pdf`.

4.3 Implementation libraries surveyed

Short list of candidate Python libraries considered for each major step, with the choice and rationale.

5. Assumptions

5.1 Modelling assumptions

Stationarity, independence, linearity, Gaussianity, etc. State each, mark its role, and rate confidence.

5.2 Data assumptions

Schema, missingness, error structure, time ordering, etc.

5.3 Computational assumptions

Numerical precision, convergence tolerances, random seeds, parallelism.

5.4 Justification table

Label	Assumption	Justification	Conf.
A1	...	...	*H/M/L*
A2	...	...	*H/M/L*
A3	...	...	*H/M/L*
A4	...	...	*H/M/L*

6. Notation & Definitions

6.1 Symbols

Symbol	Meaning
n, N	*sample / population sizes*
$\boldsymbol{x}_i$	*feature vector for unit i*
$\boldsymbol{y}$	*response vector*
S_{ij}	*pairwise similarity*
α	*significance threshold*
$\hat{\theta}$	*parameter estimate*

6.2 Domain definitions

Define every domain term the problem uses (and the team uses) before its first appearance in the analysis sections.

6.3 Statistical conventions

Hypothesis test conventions, multiple-comparison corrections (Bonferroni, BH, FWER), confidence interval method, bootstrap conventions.

7. Data

7.1 Source and provenance

Where the data came from, when it was supplied, in what format.

7.2 Schema

Column-by-column description with types, ranges, units.

7.3 Preprocessing

Cleaning, imputation, normalisation, deduplication — every step that could change a downstream result.

7.4 Quality and integrity checks

Row counts, hash, null patterns, outlier inventory.

7.5 Exploratory data summaries

Headline summary statistics, key visualisations, anything surprising before modelling started.

8. Methodology Framework

8.1 Pipeline overview

End-to-end diagram or bullet list: ingest $\rightarrow$ *preprocess* $\rightarrow$ $model_k$ $\rightarrow$ *validate* $\rightarrow$ *report.*

8.2 Software stack

Python version, key libraries with versions, OS, hardware.

8.3 Reproducibility

Random seeds, deterministic operations, how to re-run from the .ipynb in one command.

8.4 Performance considerations

Where the bottleneck is, how it was handled (vectorisation, parallelism, sparse representations, etc.).

9. Task 1 — *[title]*

9.1 Objective

...

9.2 Mathematical formulation

...

9.3 Derivation

...

9.4 Algorithm and implementation

...

9.5 Validation strategy

...

9.6 Results

...

9.7 Interpretation

...

9.8 Discussion (for Q&A)

Edge cases, alternative readings, anticipated judge questions.

10. Task 2 — *[title]*

10.1 Objective

...

10.2 Mathematical formulation

...

10.3 Derivation

...

10.4 Algorithm and implementation

...

10.5 Validation strategy

...

10.6 Results

...

10.7 Interpretation

...

10.8 Discussion (for Q&A)

...

11. Task 3 — *[title]*

11.1 Objective

...

11.2 Mathematical formulation

...

11.3 Derivation

...

11.4 Algorithm and implementation

...

11.5 Validation strategy

...

11.6 Results

...

11.7 Interpretation

...

11.8 Discussion (for Q&A)

...

12. Task 4 — *[title]*

12.1 Objective

...

12.2 Mathematical formulation

...

12.3 Derivation

...

12.4 Algorithm and implementation

...

12.5 Validation strategy

...

12.6 Results

...

12.7 Interpretation

...

12.8 Discussion (for Q&A)

...

13. Task 5 — *[title]*

13.1 Objective

...

13.2 Mathematical formulation

...

13.3 Derivation

...

13.4 Algorithm and implementation

...

13.5 Validation strategy

...

13.6 Results

...

13.7 Interpretation

...

13.8 Discussion (for Q&A)

...

14. Task 6 — *[title]*

14.1 Objective

...

14.2 Mathematical formulation

...

14.3 Derivation

...

14.4 Algorithm and implementation

...

14.5 Validation strategy

...

14.6 Results

...

14.7 Interpretation

...

14.8 Discussion (for Q&A)

...

15. Task 7 — *[title]*

15.1 Objective

...

15.2 Mathematical formulation

...

15.3 Derivation

...

15.4 Algorithm and implementation

...

15.5 Validation strategy

...

15.6 Results

...

15.7 Interpretation

...

15.8 Discussion (for Q&A)

...

16. Task 8 — *[title]*

16.1 Objective

...

16.2 Mathematical formulation

...

16.3 Derivation

...

16.4 Algorithm and implementation

...

16.5 Validation strategy

...

16.6 Results

...

16.7 Interpretation

...

16.8 Discussion (for Q&A)

...

17. Cross-Task Synthesis

17.1 Shared computational primitives

Identify the matrices / graphs / estimators reused across tasks. Confirms the report is one coherent model, not eight unrelated ones.

17.2 Consistency between tasks

Where Ti and Tj touch the same quantity, do their numbers agree? If not, why?

17.3 Composite findings

Insights that no single task produces but several tasks together imply.

18. Exploratory Analyses

18.1 Alternative formulations

Each one: motivation, change, what it produced, why it was or wasn't adopted.

18.2 Robustness checks

Vary inputs and assumptions; record what stayed stable.

18.3 Hypothesis exploration

Non-prescribed questions you asked of the data.

18.4 Counterfactual scenarios

What would the answer have been if some condition were different?

18.5 Failure mode analysis

Cases where the model gives an absurd or unstable answer — and what the team would do about them.

18.6 Findings from chasing the data

Surprises, anomalies, sub-stories worth telling in the oral presentation but cut from the 10-page report.

19. Sensitivity & Stability Analysis

19.1 Parameter perturbation

...

19.2 Threshold sweeps

...

19.3 Bootstrap and resampling stability

...

19.4 Numerical-stability checks

...

20. Validation & Verification

20.1 Synthetic data validation

Generate data with known ground truth; show the method recovers it.

20.2 Cross-method agreement

Where two methods can attack the same task, run both and compare.

20.3 External benchmarks

If applicable: a published dataset / known answer the method should reproduce.

21. Limitations

Honest list. Each limitation gets one or two sentences and (if any) the mitigating evidence.

22. Strengths

What this approach does that an obvious baseline does not.

23. Conclusions

What the eight tasks taken together answer; what the exploratory work added.

24. Recommendations & Future Work

If we had another week / month / dataset, what would we do?

25. References

[1] *Author, Title, Venue, Year. URL.*

[2] ...

[3] ...

A. Full Mathematical Derivations

Every step of every derivation hinted at in the body.

B. Algorithms & Pseudocode

Each algorithm in pseudo-code with complexity analysis.

C. Data Tables

Full result tables (the body only shows summaries).

D. Additional Figures

Plots that did not fit in the body.

E. Selected Code Listings

```
# placeholder: paste the most diagnostic snippets here
def example():
    return None
```

F. AI Prompt Log Summary

Optional: a summary index pointing to the full prompt-log PDF that is uploaded alongside the report.